

Andrew Yuan

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Technical Skills

Languages: Java, PHP, Python

Backend: MySQL, PostgreSQL, Redis, DynamoDB, Elasticsearch, RESTful APIs, Microservices, Spring Boot

Frontend: JavaScript, React, Vue

DevOps & Cloud: Azure, AWS, Docker, Kubernetes, CI/CD Pipelines

Testing: Agile/Kanban, Software Development Lifecycle, Performance Optimization, Communication Skills

Summary

Backend Software Engineer with 4+ years of experience designing and building scalable, secure, and DDD systems. Experienced in object-oriented programming, RESTful API design, and database-driven architectures. Experienced in Azure, AWS, SQL optimization, and high-concurrency services. Practical experience with AI-driven MCP (multi-channel processing) projects and AI agent development. Active open source contributor.

Experience

Uniuni Vancouver, BC
Software Developer July 2025- Nov 2025

- Contributed to a package sorting system handling over 500,000 orders daily, optimizing the order transit state process, and implementing a package tag system.
- Designed and implemented a scalable shipment tag management system using Laravel, leveraging event-driven architecture (Events/Listeners), distributed locking for concurrency control, repository pattern, and action-based tag.
- Optimized SQL query APIs by refactoring query logic and improving database indexing, reducing response time from approximately 1 minute to around 6 seconds.

TBoostAI Vancouver, BC
software engineer March 2025 - Aug 2025
Technology stack: **JavaScript, React, Java, Python, Milvus, Azure, RAG, LangChain**

- Participated in the development of an AI-powered automotive MCP, helping to recommend the most suitable used cars and dealers to users based on key profile information.
- Integrated the **Milvus** vector database using **Java** clients to enable semantic search and vector-based data handling, and utilized Cursor to improve coding efficiency.
- Designed and implemented a multi-stage state machine AI agent using **Springboot**. Integrated SMS (**Twilio**) and email sending functionalities.

China Unicom Xi'an Software Research Institute
Software Engineer May 2021 - Dec 2023
Tech Stack: Java, React, React Native, Redis, Kafka, Doris, Elasticsearch, AWS S3, Hibernate, MySQL, Docker, Kubernetes, DDD

- Led in the development of the 10010 Hotline product-recommendation module serving hundreds of millions of users: optimized **Redis** operations with **Lua**, utilized asynchronous and **strategy design pattern** to refactor the existing module, stress-tested to sustain **80,000 QPS** for this module, and boosted response speed by 5%. Reached the **TPS 5000**.
- Architected and optimized the China Unicom data visualization dashboard using **lazy rendering, front-end caching, SQL optimization, and result caching**, storing 20 GB of data on **Doris**, and improved query rendering time from 2 seconds to 200 ms.
- Utilized **XML-based BPMN** to design and automate the report approval process using the Activiti workflow engine, Streamlined the process through **OA notifications** and **SMS reminders**. Reduced manual processing time by 30%.
- Implemented **WebFlux-based survey sending + callback APIs** for high-volume questionnaire campaigns, and integrated **Netty SMS gateway** to handle message delivery, status callbacks, and user replies in real time.
- Built a Kafka-driven work-order warning system: ingested **order messages** via **Kafka** and indexed logs in **Elasticsearch (ELK)** for real-time monitoring and analysis.

ShunFeng Technology(Global 500)

Full Stack Engineer

Tech Stack: Java, VUE, Redis, Kafka, Echart, Caffeine

- Participated in a cash management system, enabling cash pooling, real-time account monitoring, and automated financial reporting to improve enterprise-wide fund visibility and efficiency.
- Java for performance-critical backend modules, with Lua used primarily in Redis scripting.
- Integrated a map drill-down feature using the **ECharts** framework to display financial data for each small region.
- Utilized **Vue** frameworks with core plugins such as **Vue-router**, **Vuex**, **Axios**, **element-ui**, **echarts**, and **vue_awesome** to build the cash management system front end.

May 2020 - May 2021

Education

New York Institute of Technology-Vancouver

Cybersecurity

Master Degree • 3.94/4.0
Jan 2024 - Dec 2024

Washington State University

Computer Science(Artificial Intelligent Track)

Master Degree
Sep 2017 - Jan 2020

Jilin University of Finance and Economy

Computer Science

Bachelor degree
Sep 2013- Jul 2017